

Wesley Rancher

Department of Geography

University of Oregon

Terrestrial Ecosystems Ecology and Landscapes Lab

wesr@uoregon.edu | [Personal Website](#)

Summary

Environmental data scientist with expertise in remote sensing, GIS, and landscape ecology. Experienced in applying machine learning to large geospatial datasets to track ecosystem change, with a focus on biomass dynamics and disturbance. Passionate about using data-driven insights to support environmental management and inform decisions that connect natural and built systems.

Education

University of Oregon | M.S. in Geography, GPA: 3.9

Ohio Wesleyan University | B.A. in Environmental Studies and Geography, GPA: 3.5

Skills

- Earth systems analysis, geospatial data processing, ecological modeling, and applied machine learning
- Proficient in Python, R, and geospatial/ML libraries (GDAL, Rasterio, GeoPandas, scikit-learn, tidymodels, randomForest)
- Clear technical writing and strong presentation of complex scientific or spatial data
- Effective coordination across scientific, engineering, and stakeholder groups
- Provides steady peer guidance and maintains reliable, well-documented, reproducible workflows

Professional Experience

GIS Analyst – American Electric Power

October 2025 – Present

- Produce cartographic exhibits and spatial analyses supporting land acquisition and utility infrastructure planning
- Develop automation tools with ArcPy to convert CAD engineering drawings into georeferenced GIS features for integration into enterprise geodatabases
- Apply optical character recognition (OCR) using Python libraries to parse geospatial information from historical property deeds to reconstruct parcel boundaries and points of reference
- Assist with preliminary site suitability analysis for substation relocation by evaluating parcel acreage, proximity to transmission lines, land cover, and topography
- Maintain and query enterprise geospatial databases supporting real estate and infrastructure planning workflows across six US States

Graduate Research Assistant – University of Oregon

September 2023 – Present

Advisor: Dr. Melissa Lucash

- Collaborated with Bonanza Creek LTER to analyze forest change and wildfire impacts in boreal ecosystems
- Conducted change detection of aboveground biomass from 2000–2025 across 8 million hectares in interior Alaska by integrating Landsat time series with random forest models and the LANDIS-II process model
- Developed Landsat preprocessing workflows in Google Earth Engine, including atmospheric and topographic correction and cross-sensor calibration following established methods
- Built reproducible ecological simulations by containerizing LANDIS-II with Docker and integrating large climate datasets (CMIP5/6) to improve biomass modeling in the Oregon Coast Range in collaboration with the U.S. Forest Service

Graduate Research Fellow – NASA DEVELOP

June 2023 – August 2023

Advisor: Dr. Anthony Vorster

- Mapped invasive plant communities in Utah's Paria River Watershed for Grand Staircase Escalante Partners using remote sensing and field data
- Integrated vegetation indices, senescence metrics, and field observations with random forest models to predict plant cover
- Processed LiDAR and Landsat datasets with ArcGIS and Google Earth Engine, creating reproducible workflows for multi-scale spatial analyses
- Co-authored a technical report and delivered virtual presentations to partners and stakeholders, translating complex geospatial results into actionable insights.

Undergraduate Research Assistant – Ohio Wesleyan University

December 2022 – May 2023

Advisor: Dr. Nathan Rowley

- Modeled supraglacial lake depth in Western Greenland using radiative transfer equations
- Built reproducible Landsat workflows in R for raster sieving and lake delineation from NDWI
- Calibrated dual-red-edge (Micasense) and LiDAR sensors on DJI drones
- Prepared educational material for FAA Part 107 exam

Undergraduate Research Fellow – University of Central Oklahoma

June 2022 – July 2022

Advisor: Dr. Victor Gonzalez

- Participated in NSF-funded REU analyzing climate stressors on heat tolerance of honeybees and sweat bees in Lesvos, Greece
- Built testing apparatuses, conducted fieldwork, and contributed to methodology for temperature acclimation, starvation, and thermal limit assays
- Found bees maintain heat tolerance following desiccation and starvation

Teaching Experience**Graduate Teaching Assistant – University of Oregon**

September 2023 – Spring 2025

- **Geography 485/585: Remote Sensing I** Winter 2025, Fall 2024
 - Developed lab exercises, taught GIS and remote sensing software (ArcGIS, QGIS, R), and led hands-on demonstrations to undergraduates and graduates
- **Geography 199: Global Wildfire** Spring 2024
 - Supported curriculum development, provided supplemental instruction, and assisted with student questions on wildfire concepts
 - **Guest lecture: "Changing Wildfire in Brazil"** – discussed landscape drivers of a changing fire regime in Brazil
 - **Guest lecture: "Bees and Wildfire"** – explained the interplay between post-wildfire effects, vegetation, and pollinators
- **Geography 181: Our Digital Earth** Winter 2024, Fall 2023
 - Facilitated labs focused on digital mapping and spatial data; helped students with ArcGIS Online and digital geography concepts

Awards and Honors

Riphey Research Grant (\$1000, UO)	2024
NASA Develop Scholarship (\$1500, SSAI)	2023
Dean's List (OWU)	Fall '22, Spring '20, '22, '23
Robert E. Shanklin Distinguished Scholar (Geography, OWU)	2023
Phi Sigma Tau (Philosophy, OWU)	2023
Our New Gold Digital Storytelling <u>winner</u> (Spanish, OWU)	2022

Publications

- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). Estimating recent changes in tree biomass in interior Alaska by combining Landsat imagery and random forests. *(In preparation)*
- Lucash, M., Lamping, J., Nowell, B., Scheller, R., Banerjee, T., Buettner, C., Fawcett, J., Hurteau, M., Parks, S., Rancher, W., Robbins, Z., St. Denis, L., Stasiewicz, A., Urza, A., & Weiss, S. (2025). *Roadmap for the future of extreme wildfire events. (In Review)*
- Weiss, S., Rancher, W., Hayes, K., Buma, B., & Lucash, M. (2024). *Wildfire dynamics under climate change in interior Alaska. (In Review)*
- Gonzalez, V., Rancher, W., Vigil, R., Garino-Heisey, I., Oyen, K., Tscheulin, T., Petanidou, T., Hranitz, J., & Barthell, J. (2024). *Bees remain heat tolerant after acute exposure to desiccation and starvation. Journal of Experimental Biology*
- Rowley, N., Rancher, W., & Karmosky, C. (2024). *Comparison of multiple methods for supraglacial melt-lake volume estimation in western Greenland during the 2021 summer melt season. Glaciers*

Presentations

- Rancher, W. (2025). *Estimating species-level aboveground carbon in interior Alaska using machine learning and process-based models. University of Oregon, Eugene, OR. (Master's thesis presentation)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating recent shifts in aboveground carbon and species composition in interior Alaska using Landsat imagery and random forests. Northwest Scientific Association, Eugene, OR. (Poster)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2024). *Assessing vegetation shifts in boreal Alaska by integrating Landsat imagery with spatial modeling. American Geophysical Union, Washington, DC. (Poster)*
- Rancher, W., VanArnam, M., Kowalski, A., Anarella, T., & Vorster, A. (2023). *Mapping Russian olive and tamarisk to inform invasive species management along the Paria River, Utah. NASA DEVELOP Day, Washington, DC. (Virtual talk)*
- Rancher, W., Rowley, N. (2023). *Estimating supraglacial melt lake volume changes in west-central Greenland using multiple remote sensing methods. Ohio Wesleyan Spring Symposium, Delaware, OH. (Poster)*
- Rancher, W., Vigil, R., Garino-Heisey, I., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance. Ohio Wesleyan Connection Conference, Delaware, OH. (Poster)*
- Rancher, W., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance. IUSSI Sección Andina y del Caribe, Panama City, Panama. (Talk)*